(a) Implemented browser preprocessing

Camera or gallery file
Reject absent input or file size > 15 MiB

Decode original image
Decoded pixels occupy phone memory

Resize within next bound
160 x 120, then 120 x 90, then 80 x 60
Preserve aspect ratio; no enlargement

White canvas; grayscale pixels
Y = round(0.299R + 0.587G + 0.114B)
Write R = G = B = Y

Request JPEG encoding at next q
q = 0.60, 0.45, 0.30, 0.20
Browser codec: conceptual detail in (b)

Valid output and size <= 8192 bytes?

No: next q*

Yes

Accept first fitting JPEG
Upload binary bytes to the civilian node

*** After four failures, resize at the next bound.**
Reject if all 12 candidate encodings fail.

(b) JPEG encoding model [6], [7]

Prepare component sample blocks
Baseline illustration: 8 x 8 samples

Discrete cosine transform
Pixel values -> spatial-frequency coefficients

Quantise coefficients
Divide by quantisation steps; round

Order and represent coefficients
DC differences; zigzag AC / zero runs

Entropy coding
Huffman coding in the baseline model

JPEG byte stream
Return encoded Blob to the size check

The application supplies pixels and q.
Codec details are selected by the browser.